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A linear regression penalty estimator programme for the mitigation of shortcomings in availability based tariff scheme adopted in Indian power grid networks

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As the prediction of the cost function for power exchange between the power networks is a predominant factor for effective power operation, the power operators are all subjected to paying the penalty for the power exchange over the various grid networks favored by load encroachment. The penalty imposed for the mismatching in the overdraw and under drawn of power for the power operators are all decided by various operating constraints, which could be effectively managed by introducing a modified penalty predictor model for the power exchange between the grid networks, which witnesses the overall dynamic operating nature of various power generating units. This research paper intends to bring out a penalty estimator programme based on considering multiple variables relevant to the operating condition at different time blocks arranged in a sequence of various factorizations of power indices using the curve fitting technique. The indicated power indices from the predictor model earned from various instances could be considered to impose penalties for the individual power operators. This indicator method could fit the curve, which can be rearranged multiple times through the linear regression method to bring out an optimized power operating condition. The shortcomings of imposing penalties in a flattened manner could be avoided by adopting this linear regression-based penalty indicator programme.

Keywords Power outage, Unscheduled interchange, Penalty estimation, Power exchange, Economic power operation, Linear regression method

Abbreviations

CERC	Central electricity regulatory commission
ABT	Availability based tariff
ALO	Ant lion optimization
UI	Unscheduled interchange
MATLAB	Matrix laboratory
PIP	Penalty indicator programme
LR	Linear regression
EMS	Energy management system

The term power outage is coined from the power cut, power blackout¹ and power failure, which interrupts power. The power outage is considered based on short-term interruption, the voltage drop in the transmission system –Brownouts², and the total failure of the system lasting for a few hours to a few days or a few weeks, which is considered as blackouts. The effect of this power outage is described as a power failure, which could be recovered to the normal operating state after some period of time. The real event occurs behind this failure,

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lasting from a few minutes to a few hours, which can lead to network shutting and cascading failure³ of the power components such as relays, circuit breakers and other Protective devices.

The short-term interruption of power could be resilient, which might not be considered until the system is subjected to significant failure. The outcome of power failure which lasts for over 40 hours in a widespread region of the world 2nd largest populated landscape will be a severe case to make a study on it. As the power outage on that particular instance been favored by various power mismanagement and failure of operating entities, Unscheduled interchange (UI) of power between various power generating grid networks favors the curtailment of power for an extended period of time. On making a detailed enquiry on the reasons behind the power outage events, it has been found out that drawl and under-drawl⁵ of the power in between the power dispatching centers plays a vital role in the power outset and it has been considered as the one of the primary reason for the outcome of cascaded power outages prolonged for continuous two days of the northern part of India on July 30 and July 31, 2012. As unscheduled interchange of power might be practiced between five grids sectionalized in India, the Central Electricity Regulatory Commission (CERC) has articulated the availability-based tariff (ABT) method⁶ for the transfer of power between two interstate grid regions. The ABT method has been obliged to fulfill the credible power transfer between power operators, which has been adopted in India's central and state-level grid networks. The ABT method covers three major components for up scaling the proper power interchange during outage instances. (i) system capacity charges, (ii) charge for the energy generated, and (iii) UI charges. Despite having various attributes that serve to compensate for unscheduled interchange of power in most occurrences, this method is very much restricted to consider the sudden variation of frequency and the influence of available renewable energy resources for power generation. If such additional key factors were considered for evaluating the ABT penalty indices, it would give more benefits to the power operators.

Literature review

The research article proposed by (abhishek sainsi 2022) by modifying the ant lion optimization algorithm to a improved ant lion optimization algorithm for validating the operating cost of the thermal units connected to the power system with standard IEEE 30 bus system. The research paper defends the proposed IALO which shows better results compared to other optimization tools by minimizing the objective function such as equality and inequality constraints in an acceptable range. Also the research speaks about the improvement in operating cost, active power loss and voltage profile in a given power network. The paper provides lesser information about considering the dynamic power transfer capability of thermal power units.

An optimization model for operational decision making process for the domestic consumer by setting a predetermined dynamic energy tariff scheme was proposed (Michiel Avau 2020) to address the model for distribution tariff scheme by the power provider can make the optimal allocation of power by cost efficient scheduling scheme which minimize the cost function and also the participation of the individual prosumer to participate in the dynamic power tariff. The paper introduces the relative flexibility value for lowering the energy cost based on the minimal energy consumption of the customer. The price elasticity of the demand for customers with inflexible demand, which could influence dynamic tariff structures has not been discussed in depth in this research article.

The immediate requirement of ancillary services provided by the distributed energy resources have been estimated by hierarchical coordinated automatic generation scheme proposed by (Yukang shen 2025) for the multi grid networks. In this research paper, evaluation was made of each distribution network provided that acts as an AGC entity and the regulation capacity has been estimated using an aggregation dynamic model, while the constraints in the optimization program are estimated based on curve –fitting, which can predict power response to voltage and frequency variations. But the specific setting of active constraints has not been widely discussed in this paper.

An attempt made to incorporate wind power in an interconnected power system which sensitize the frequency deviation and tie line power deviation which influence the variation in the tariff structure under the availability based tariff mechanism was proposed by (Gargi konar 2022) which makes the wind power input to be the critical factor for restricting the unscheduled interchange of power with respect to the primary and secondary frequency control of the wind turbine generator. The result outcome depicts the reduction of power generation and power cost with respect to the increase in the wind penetration keeping the frequency under control.

A penalty based secured power system model has been proposed (Minjing yang 2020) for witnessing the false data injection to the power network with the estimation of penalty based on the boundary error and dynamic internal error. A measurement model has been developed through lagrange function and compared with the state estimation algorithm and the penalty factor can be induced to improve the computational efficiency. Even though the penalty based scheme introduced, it lacks to bring up linear scale prediction of penalty imposition for the user, whenever false data injection added to the power module.

Problem statement

The above referred research articles comprises the techniques carried out to monitor the UI exchange and formulating new intelligent solution techniques¹² for bringing an adaptive power exchange mechanism. Also they have put forward some penalty estimation schemes for the violation happens during the UI power exchange. The instigation for bringing out a solution technique for regulating the availability based tariff mechanism in the UI evolved due to the outcome of cascaded outage events which stall the continuity of power over 40 hours on two consecutive days of July 2012. Though various literatures propose multiple formulation mechanisms to modulate the unscheduled interchange of power between load dispatch centers, most of the researchers resolute on fixing penalty for frequency deviation, under drawl and overdraw of power exchange between tie links of power system area. They least bother about the critical factors which will be tethered with UI phenomenon. Ignoring those

factors will certainly bringing up irrational tariff obtruded to the power generating units. This will lead to the erratic behavior of grid integration and also affect the load forecasting¹³ and load Prioritization. As the dynamic power operation of the grid network influence the economic equilibrium which has to be validated by several instances of frequency dips and rise, voltage profile during the unscheduled interchange of power for extended time period has not been considered in most of the formularize techniques which has to consider in this research paper. This research work mainly focuses on interlinking existing operating variables which are all witnessed by variation in power demand with hidden factors such as over shoot and under shoot of voltage profile in a given generating networks delivering power throughout the transmission facilities provided for bringing the linear solution for penalty estimator. Here we are considering the outage events prevails during the 2012 blackouts shown in Fig. 1 and a power quantities relevant to the failure transmission link region will be subjected to make a fitted penalty estimator technique by carried out linear regression method of curve fitting technique.

This research work intends to comprise two major factors which have not been considered on previous methodologies by relating frequency index as shown in Fig. 2 for the cost fix up in the UI mechanisms.

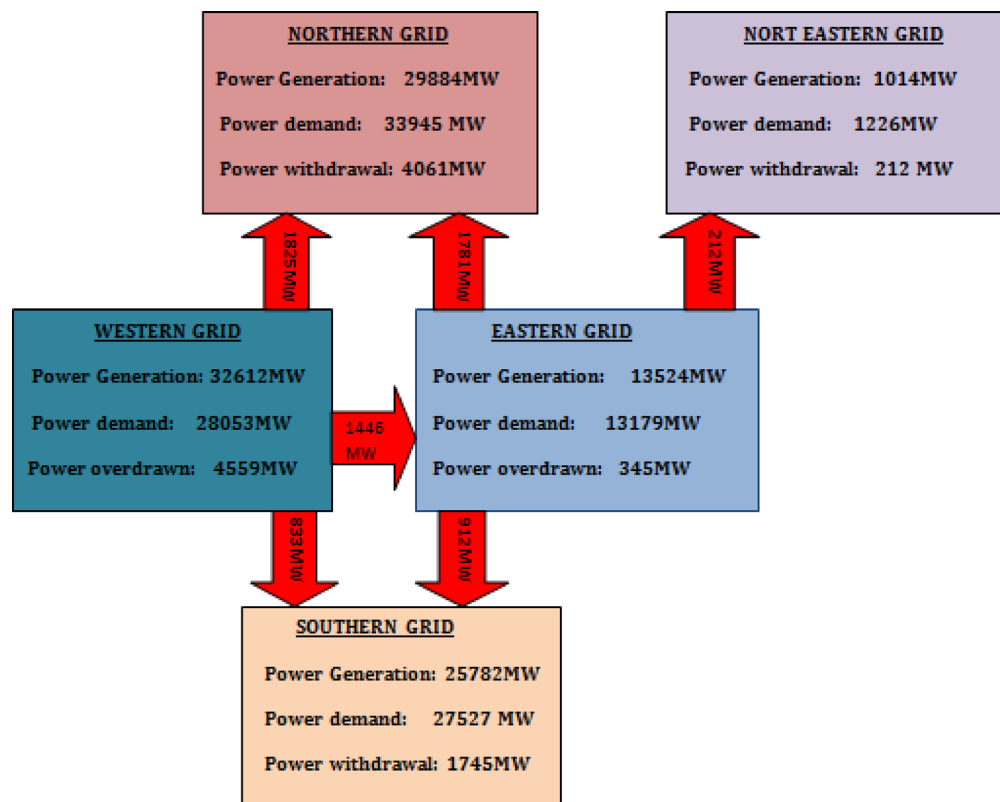
The two major restraints included in this research work are.

Restoration of the feeder after the nominal frequency reach

The power lines associated with the interlinked power exchange can be able to generate power as they supposed to generate. Different energy resources starts to generate power up to the specified capacity prescribed. In the event of 2012 power blackout most of the generating units restore the generation as per the generation schedule without considering the frequency reach of normal set value 49.5–50 Hz. In most cases the under frequency relays adopted for the protection does not take account of instantaneous fluctuation of frequency due to load imbalance state¹⁴. It is desired to observe the variation in frequency for some prolonged time period by the power supplying agencies to make a distinct decision on individual system subjected for penalty.

Decision on permitted frequency range based load shedding

As it is a mandatory rule of opting load shedding for the frequency deviation in the system performance, the centralized control mechanism which considered the frequency range of specified level 50 Hz as the prescribed set value for continuous power operation, they intends to switch off the load if the frequency range downscales from 50 Hz to even 49.5 Hz. Generally both event based and response based signals initiates the load shedding of the system, but in certain operating systems the grid operators make the decision of enforcing the load shedding based on the current situation and system requirement. This problem witnesses the flat rate fixing of penalty for which the operating frequency got deviates from its range.



Power exchange condition between grids prior to outage on 31st July 2012

Fig. 1. power exchange condition prior to power outage on 31 st July2012.

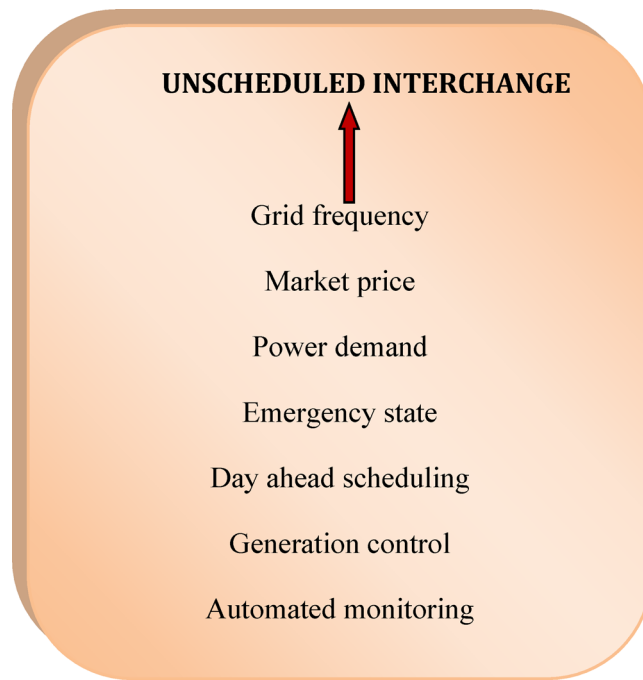


Fig. 2. Factors incorporated with unscheduled interchange of power.

Proposed methodology

The proposed penalty indicator programme (PIP) intends to provide an indicative signal for imposing penalty for the unscheduled interchange of power through the ABT mechanism¹⁶. This PIP will try to witness the follow through of frequency dips and rise throughout the power generation of the system. As existing ABT methodologies obtrude penalty based upon the frequency deviation without considering any other compelling factors which supports intended rise or lowering of frequency in the network given. This programme will be tested on a power network of 400KV range interconnected with 23 interlinking units. This PIP has considered the following criterion for setting cost condition.

Frequency deviation index

In the case of the power generating system we have considered the frequency variation stands out in most of the generating units, leaving certain generating units exhibited continuous fluctuation of frequency outside the range of 49.5–50.5 Hz. Here we are considering the certain systems imply variation of frequency outside the stated range, and the deviation count will be

$$\Delta f_1 X_1 + \Delta f_2 X_2 + \dots \Delta f_n X_n \quad (1)$$

of all the 'n' number of units provided variation of frequency change outside the limit with respect to cost functions X, certain units involves extended frequency deviation gets shortlisted¹⁷ and the frequency deviation index will be framed out from the Eq. 1 as

$$fk_d = \Delta f_d X_n t_d \quad (2)$$

the generating functionality fk_d is given by the product of deviation in frequency from the nominal range Δf_d , number of strikes the frequency range¹⁸ far off the nominal range X_n and the time count t_d of the deviation prevails for the particular generating units.

Overdraw and under draw of power within the grid region

During the 2012 blackout occurrence the generating units tends to overdraw power from the eastern grid region and subsequently prevails under drawl of power in most of the eastern grid generating units of the country. Our research work aims to include the veering of power exchange between the interconnecting systems as.

Amount of excessive power withdrawn by the generating stations,

$$p_w = \sum_{k=0}^n pwi \quad (3)$$

Amount of under drawn power by the generating stations,

$$p_u = \sum_{k=0}^n pui \quad (4)$$

Total amount of power variation in between the units is considered as

$$p_1 = \sum_{k=0}^n pwi = \sum_{k=0}^n pui \quad (5)$$

The sum up of power in over drawl and under drawl in addition to the various other transmission constraints taken into account of programme for the decision making of regulated penalty indicator scheme whereas.

Additional transmission constraints taken into account as X_s for the generating units $k = 1, 2, \dots, n$

Enforced violation on power exchange operation

Due to various powers mismatch condition, operating failure, protective device dysfunctional cases there exist the possibility of breaching the power exchange limit¹⁹ assigned for the UI exchange. This regulated penalty indicator programme take into an account of such instances and get assigned in the proposed PIP as shown in the Fig. 3.

$$P_0 = T_d = ktr + f_r + \Phi_a + \Phi_g \quad (6)$$

provided, $T_d > 0$ for the violation exceeds the limit.

$T_d = 0$ for violation free power exchange.

Where T_d – transmission violation coefficients.

k_{tr} - transmitted power coefficient.

f_r - frequency range.

Φ_a - damper coefficients.

Φ_g - generator heat coefficients.

Linear regression penalty indicator programme

For identifying the penalty margin for the n number of generating units connected in the grid section out of all decisive factors involved, the above mentioned operating constraints get integrated into a linear formulate solution for making decision for impose penalty for the power generating operator. This penalty indicator programme will be performed based on equating all the three contributing factors from Eq. 2, 5, 6 and create the factorization of power drawl condition for the n generating units integrated in the power transmission tie line. Based on the three factors pertained to each other the decision of penalty imposition will be made.

Test case for penalty indicator programme

For the evaluation of frequency mismatch and violation of unscheduled interchange of power in the ABT scheme the transmission line of 440 KV Gwalior-Bina tie line of 450 km in the northern grid region has taken for the test case and the power exchange of 23 generating units in that tie line been considered and the penalty evaluation

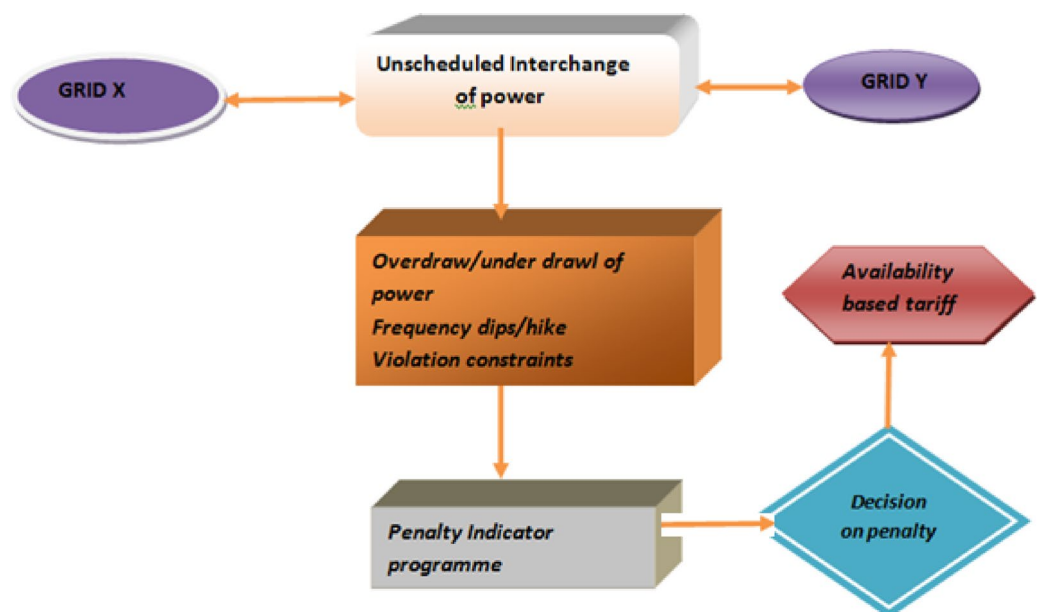


Fig. 3. Flow representation of proposed penalty indicator programme.

by the proposed penalty indicator programme got compared with the existing ABT mechanism²⁰ which follows only frequency variation as the considerable factor.

On the 400 KV Bina-Gwalior line subjected to the flow of 1000 MW power on July 30th 2012, which plays a key role for the huge power outage befall on 30th July 2012, our PIP intends to take all operating details of the generating units comprises in the northern and eastern grid section, which already been penalized for the overdraw of power at the sudden failure of the system. Apart from the decision of penalty inflicted on the certain system, the proposed integrated PIP ensures the consideration of other three substantial factors that might influence the penalty decision²¹ for the UI exchange in between the power line.

Implication of linear regression method for PIP

For the detailed analysis of a data without any information, one can use this regression method by having the value of an independent variable to obtain the data of a dependent variable which is the linear regression (LR) method of evaluating the unknown variable by formulating all the known variables in a linear equation form which is shown in Fig. 4. As LR method is the suited one for developing predictor models by associating both dependent and independent variables, our research intended to make use of the LR method by correlating the specified power quantities with unspecified cost function to estimate the penalty for individual system connected to the grid.

Linear regression method for penalty estimation	
Dependent variable	Generator coefficients
Independent variables	Power, frequency & transmission coefficients
Power data	Under/over drawl of power
Slope of regression	Penalty free power exchange range

Factorization and evaluation process

Though LR method implies straight slope for predicting the dependent variables, in certain cases linear curve fitting can be defensible one for our estimation process. The set of variables considered for the curve fitting condition were polynomials²² of row vectors contains coefficients which we are all denoted in descending powers.

For the consideration of our research work.

The general linear regression condition can be given as.

$Y_i = b_o + b_1 X_i$ can be fit into the power function as,

$$P = [P_2 P_1 P_0] \tag{7}$$

Where P_2, P_1, P_0 relates the power range over drawn and withdrawn from the neighbor network.

The integrated power relation with respect to frequency index and other violation constraints will be formulated as a second order linear equation²³ given as

$$P(x) = p_2x^2 + p_1x + p_0 \tag{8}$$

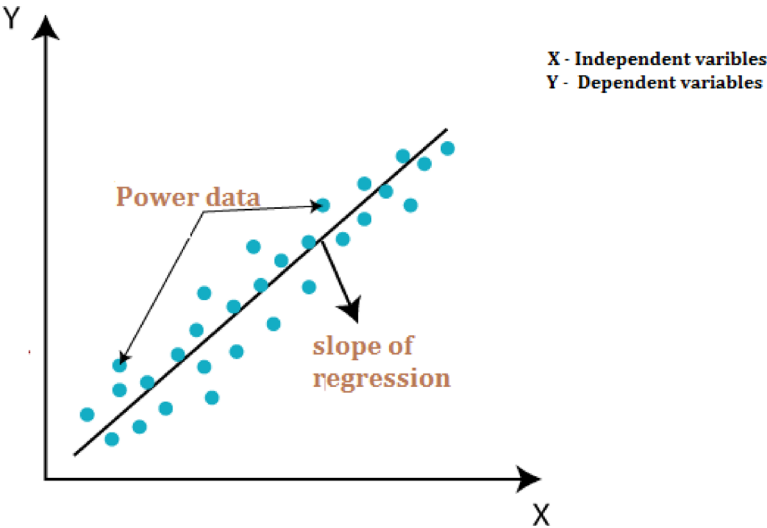


Fig. 4. Linear regression model for penalty estimation.

$$Y(x, p) = p_o + p_1x + p_2x^2 + \dots + p_mx^M \sum_J^M P_j X_j \quad (9)$$

j – number of iterations carried out for the various power instances.

M- number of power units subjected to the power interchange exercise.

With the linear condition obtained for the power exchange between the generating indices the polynomial derivative for the n number of units interconnected together will be obtained by the programme objective function as

$$P_n = \text{Polyfit}(x, y, n) \quad (10)$$

by which,

- x and y vectors containing the x and y coordinates of the data points.
- n represents degree of the polynomial to fit.
- p is the vector corresponds to polynomial coefficients.

Create some x-y test data for five data points.

By using the polyfit to find a third-degree polynomial that approximately fits the data, the above power ranges varied will be considered for the polynomial set of variables with respect to various constraints. we have taken into consideration for making the decision of penalty for the operators using the polynomial curve fitting estimation method from the above condition set gets addressed eventually with the system operation for the period of t with n intervals which can be evaluated to decide the UI tariff. On the basis of obtaining various operating points in the polynomial curve the cost function for the generating units can be estimated²⁴.

The selection procedure for the polynomial curve fitting programme from the obtained power quantities can be performed by the following steps as shown in Fig. 5,

1. Consideration of power entities in the units interconnected over the grid section.
2. Evaluation of power exchange indices and the scheduled power rating.

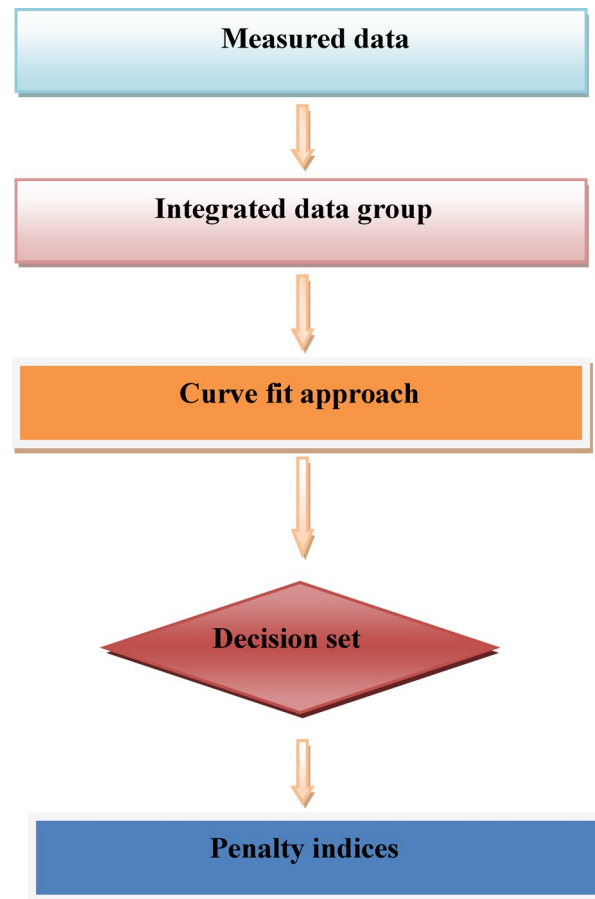


Fig. 5. Cost estimation process through curve fitting approach.

Station details	Actual injection of power (in MWh)	Scheduled generation of power (in MW)	Deviation of power (in MW)	Deviation nature	Deviation rate in percentage	Penalty rate in Rs/KWh
Singrauli	1545	1500	45	Over drawn	3%	53,766
Rihand I&II	1403	1390	13	Over drawn	0.9%	52,200
Unchahar	844	850	-6	Under drawn	0.7%	57,900
Dadri-I&II	1463	1380	83	Over drawn	0.6%	52,200
Jhajjar	415	415	0	No deviation	-	-
Shree cement	197	267	-70	Under drawn	26%	1,78,300
Dadri gas	548	538	10	Over drawn	0.1%	27,900
Auraiya gas	306	302	4	Over drawn	0.1%	27,900
Anta gas	354	349	5	Over drawn	0.1%	-
Narora	236	258	-22	Under drawn	0.8%	53,900
Nathpa jhakri	411	507	-96	Under drawn	19%	62,218
Bhakra	637	656	-19	Under drawn	0.2%	27,900
Tehri	753	760	-7	Under drawn	0.9%	58,300
Dehar	600	573	-27	Under drawn	0.4%	34,900
Chamera I	365	356	-9	Under drawn	0.2%	27,900
Salal I&II	660	601	-59	Under drawn	1%	74,900
Uri	432	430	2	Over drawn	0.4%	34,200
Pong	108	108	0	No deviation	0	-
Dulhastri	390	390	0	No deviation	0	-
Dhauliganga	282	284	-2	Under drawn	0.7%	57,900
Chamera	302	302	0	No deviation	-	-
Tanakpur	94	93	1	Over drawn	1%	74,900

Table 1. Power deviation details of Northern grid region on the outage event of July 2012.

Frequency range	Recommended penalty for power exchange variation
Below 49.5 Hz	870 Paise/Kwh
49.5–49.7 Hz	415 Paise/Kwh
50–50.2 Hz	215 Paise/Kwh
49.8–50 Hz	210 Paise/Kwh
Above 50.5 Hz	870 Paise/Kwh

Table 2. Penalty rate for the frequency deviation.

3. Taking account of other external factors contribute to the excessive and modest value of power between the connected power units.
4. Equating all the above functions in a polynomial cost derivative function.
5. Making use of polyfit to the variables.
6. Equating the linear condition fit the curve to the indicated margin in the x, y coordinates.
7. Based on the evaluation appropriate decision on penalty can be made on power entities.

Consideration and evaluation of power exchange deviation in the power network can be recorded by having the deviation of power²⁵ in time block of the entire operation.

Time block regulated as per Central electricity regulatory commission (CERC) recommendation: a twenty four hour day period has been divided into various time blocks, whereas each time block of 15 minutes.

Penalty indicator programme variables: $P = [P_2 \ P_1 \ P_0]$ which includes external factors enforce the limit violations.

As per CERC guidelines estimated UI cost is 870 paise/kwh.

As per the PIP, for 24 h operation the total time block will be: $24 \times 60 = 1440$ min/day then 24 h time period will be covered by, $1440/15 = 96$ time blocks/day.

For considering the deviation rate per time block specified, the formula is given by:

For the Gwalior-Bina grid network we are considering for the formulation of PIP, the deviation index is tabled out in Table 1 as follows.

From the deviation chart of the northern region, the penalty decision can be made based on the penalty fix up for the power units. The deviation penalty cost was framed based on the recommendation made by the central electricity regulatory commission (CERC), India (Table 2) in the year of 2010.

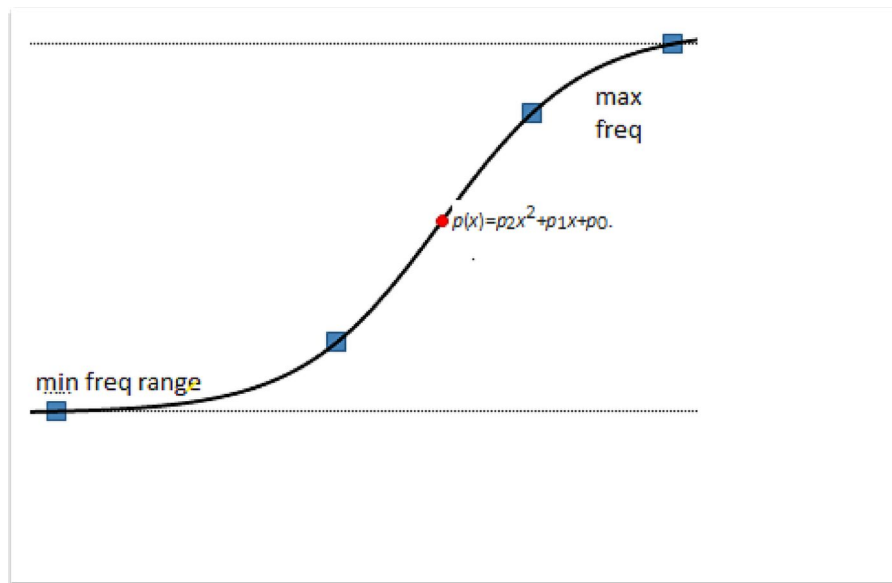


Fig. 6. Frequency fit to the curve.

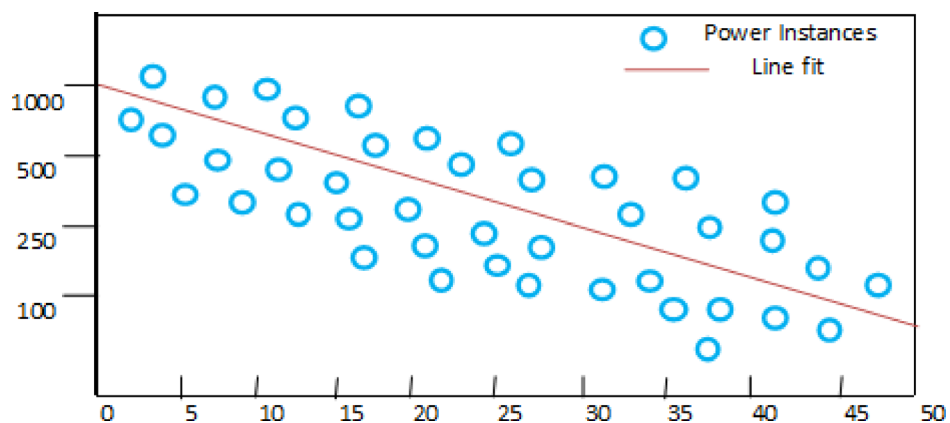


Fig. 7. curve fitting of various power indices.

Based on the recommended penalty cost shown in Table 2 setting for the group deviation of power exchange provided in the northern grid region of the Indian power system the total deviation penalty for the particular day of 30th July 2012 has been calculated as.

Average Percentage deviation for the over drawn of power: 0.8%.

Average percentage deviation for the under drawn of power: 5.2%.

Total penalty cost for the generating stations as per the guidelines prescribed will be Rs.7,30,684 per hour, which will not be a penalty imposed considering all defining aspects. It is quite proven that every generating units intended to fall out of prescribed operating condition, the conventional method of charging through the ABT mechanism will not favors the linear uniform approach of penalty. The PIP technique adopted here will make sure every system subjected to deviation will be treated on the merit of three factors endorsed in the previous section. In order to ensure the penalty derivative with equally treated condition, all the data's referred with the frequency, voltage profile and other constraints²⁶ gets fitted in the linear equation based on curve fitting technique²⁷ shown in Fig. 6 that we have adopted in this research work.

From the initial curve fit the second order polynomial function which got to be fitted within the boundary condition reference to x, y coordinates as x refers the time function and y refers the power increment. The condition gets substituted by polynomial regression technique²⁸ as such one can perform the cost function without having any linear relationship between the time function and power increment in MW plotted on X and Y axis respectively, which is provided in the data set shown in Fig. 7 and the relationship between the time function and power over drawl and under drawl range been plotted on X and Y plane respectively which is represented in Fig. 8.

As second order function got implies for the objective function the validation will be preceded by linear polynomial function²⁹.

Linear fit of data with penalty prediction interval

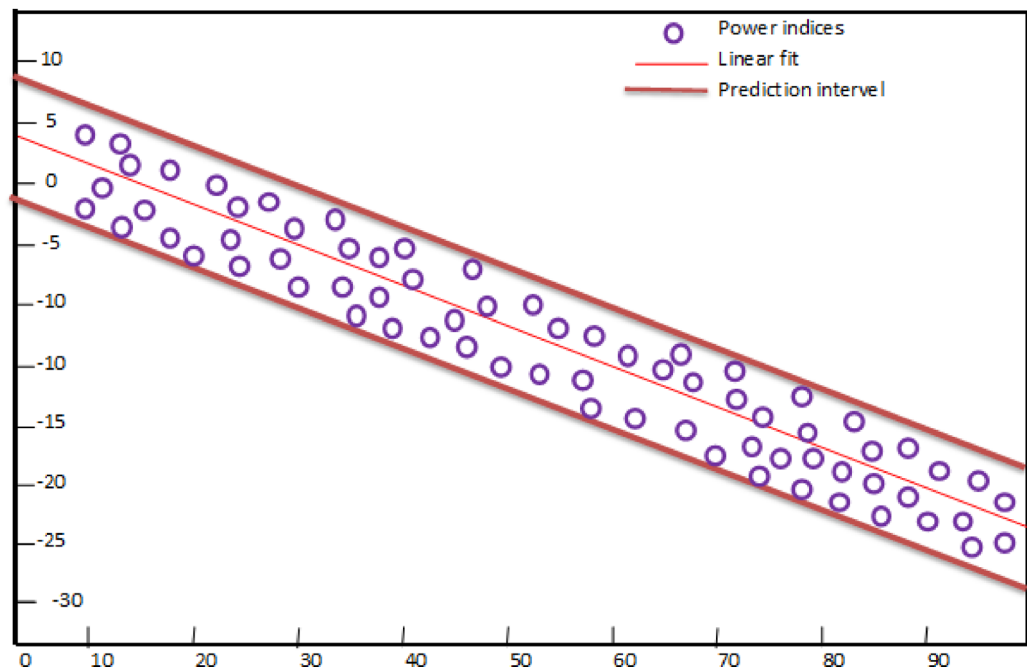


Fig. 8. Penalty prediction fitting curve.

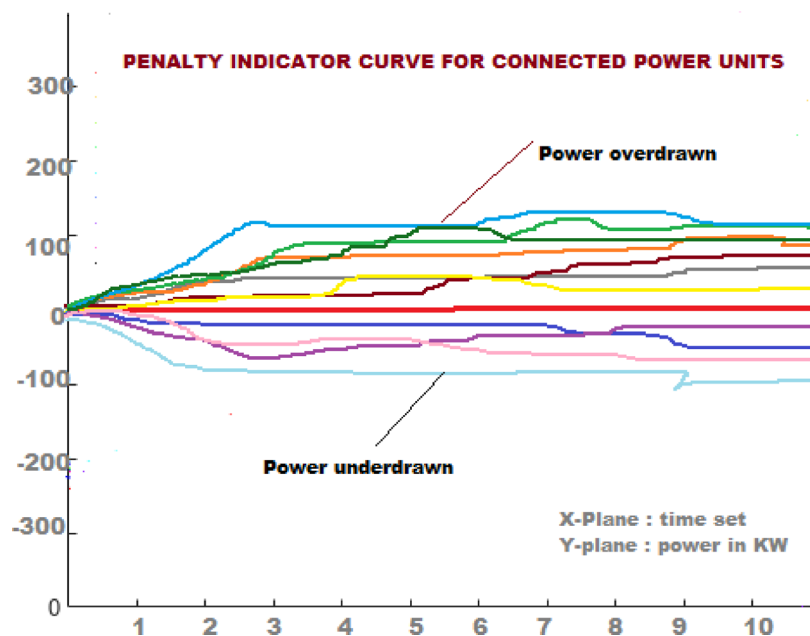


Fig. 9. power variation of different generating units.

As the time block setting for the ABT measurement is 15 minutes for each time set, total time blocks for the 24 h get factorized with all the polyfit function (x, y, n) refers above. By which we are creating a vectors of power data on (x, y) coordinates with the help of poly fit function for obtaining the second order degree polynomial to the data. By formulating the polynomial linear regression programme the linear model with the data points of frequency, voltage and other constraints are fit into the curve and the result gets plotted as shown in Fig. 9, which will give the indicator regard the penalty for under drawn and over drawl condition³⁰.

The above linear fit curve impels the energy centers to make a decision on the inflicting penalty charges over the various generating units integrated together in a dynamic operating condition³¹. Based on the magnitude of the limit violation made by the units, the penalty charging can be formulated which is based on the number

of instances the generators deviates the power limit³² which is formulated as mentioned in the Table 3 provided below.

By framing the penalty margin for the system operators in an energy management system (EMS)³³, it could be adaptable by the power providers as such specific instances of rendering flexibility for sharing the power through renewable energy sources and hybrid power utilities through all power networks³⁴.

Results and discussion

With the details of system got trapped in a power outage on July 30th 2012, the power outage data of northern grid Gwalior-Bina section got considered for the evaluation of unscheduled interchange of power beyond the permissible limit allocated by the control centre. The previous penalty estimation methodology is purely based on the ABT mechanism which follows the conventional way of considering the frequency variation over the system operation and the irregularity prevails in the power exchange³⁵ between the connected networks. The proposed PIP based on the linear regression curve fitting technique ensure the estimation of penalty cost based on the generation capacity, resource constraints, voltage profile, frequency deviation and other dynamic factors. The way considering multi constraints for setting the penalty estimator program provides much refined representation of system behavior and serves more purpose to the power operators. Also the multi staging penalty scheme brings out the substantial control of operating constraints to the dynamic power entity which avoids the suboptimal and infeasible state of the power system. As per the guidelines provided by the CERC³⁶, metering, accounting and settlement for the UI should be done with at most priority³⁷ of reducing complexities in the grid integration in the state level operation. To achieve such state of process in UI exchange our proposed methodology will pave the way of giving clear indication to the power operators to understand the power exchange mechanism done over the given entity³⁸. Based on the power exchange data obtained from the generating units of Gwalior-Bina transmission corridor, the existing penalty figure 10 obtained will get compared with the penalty data through our PIP method and the comparison outcome has been displayed in the Table 4 given below.

As the penalty index for the system is based on the UI cost of the system which is 870 paise/kWh as per the CERC norms that impose to the available system excludes systems without penalty. The decision of bringing penalty to the system purely decides the variation of power exchange values which is out of scheduled exchange. If the system violates the exchange³⁹ within the specified range of deviation that might not be considered in an existing tariff mechanism. The main reason behind this situation is based on the percentage of UI deviation rules stated that for those systems, claims less than 2% of deviation which can be exempted from the penalty block⁴⁰. In the proposed scheme of penalty indicator method the penalty estimation is based on the number of time blocks whereas the power exchange violation entertained. Also the other over drawl and under drawl of power from the particular power corridor⁴¹ been considered by linearizing several operating constraints through curve fit technique. This one leads to make a unified decision of imposing penalty for various generating units based on the operating instances⁴² which has not been considered and formulated in conventional ABT mechanism.

Conclusion

The result outcome from the above Fig. 11 shows by assimilate the PIP technique in the penalty estimation, by plotting the time block set for individual power line on X-axis and power range on Y-axis. The amount fined for power violation is evenly balanced based on the violation index of the generating units which got compared to the existing ABT mechanism. In this research work an integrated penalty indicator programme was formulated by considering frequency deviation, overdraw and withdrawal of power during UI exchange through a linear curve fitting technique. This technique will undergone a linear regression based factorization of second order polynomial function which makes the linear fit of various constraints under the boundary condition of penalty margin. Based on the programme formulation using Matlab software of linear fit regression model technique the penalty decision is made by keeping the time block as a fundamental decisive factor for imposing penalty for various generating units. by implementing this PIP method the penalty index for the power operators got improved upto 15 to 20% as per the results evolved from the taken samples of power network. As mentioned above the key motivating factor for bringing out this penalty estimation methodology is the outburst of drastic events occurs during the double-headed power outage happens in the country during July 2012, where UI of power plays a vital role in the outset of that outage event. This research work could pave the way for bringing new innovations in the automatic monitoring and modified forecasting techniques to overcome the challenges in the penalty estimation. The result shows the proposed PIP method bringing more credibility in imposing penalty to the power operators based on the real time condition prevails in the network which has been recommended by the power authorities nowadays. This work will hints out a possibility of bringing up more specific regulatory measures for eliminating the power violation around the world.

Violation count/time block	Penalty cost	Deplorable condition prevails
1–4 times	No penalty	Overdrawn/under drawl of power
5–10 times	20% of UI cost	
10–25 times	30% of UI cost	
25–50 times	50% of UI cost	
> 50 times	75% of UI cost	

Table 3. Penalty based on violation strike/time block.

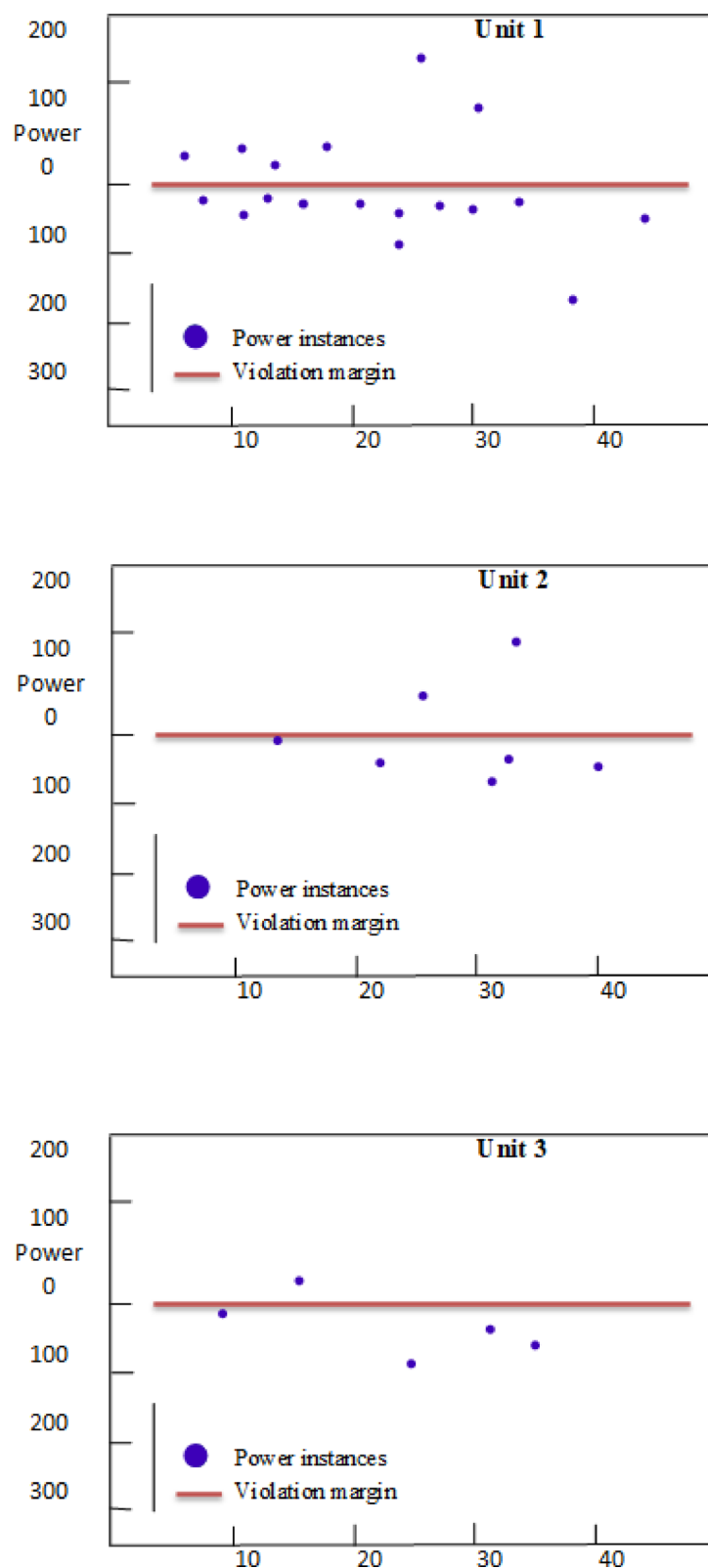


Fig. 10. simulated results showing the ten various generating units power violation instances based on PIP methodology.

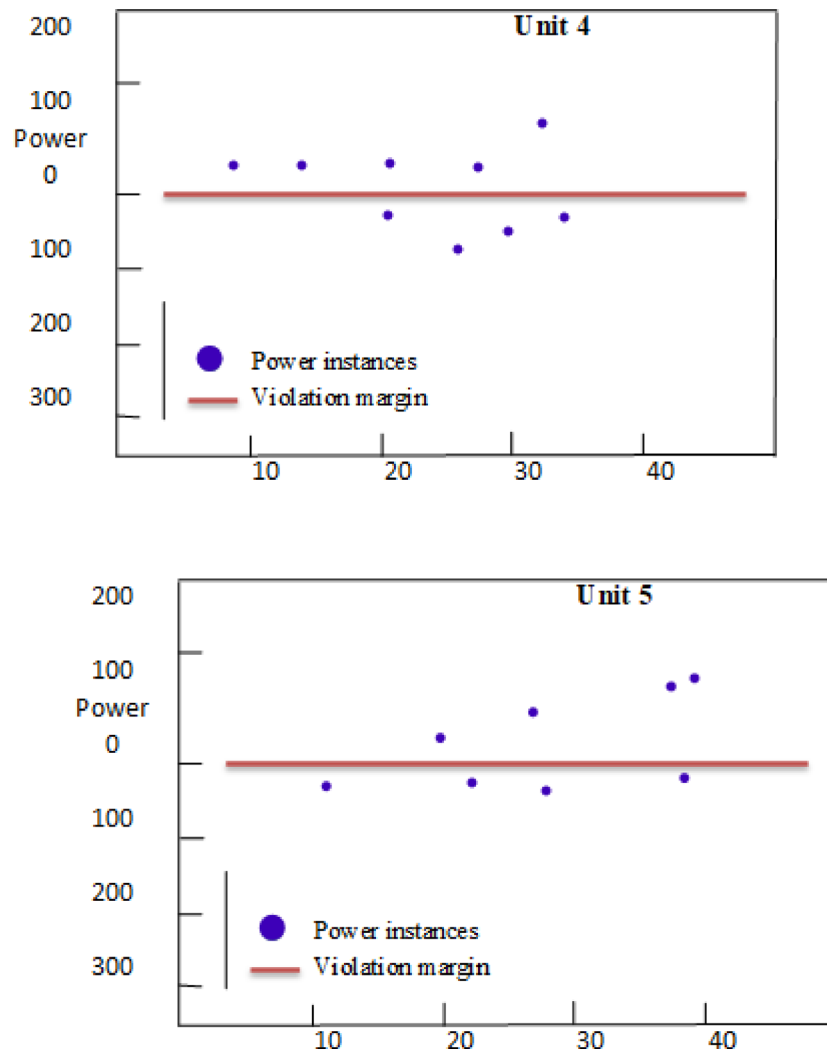


Fig. 10. (continued)

Declaration of data

As the power outage statistical data's and the cost evaluation for the unscheduled interchange between the power lines have been stated based on the report of the enquiry committee on grid disturbance submitted to the union government of India and the amendments formulated by central electricity regulatory commission respectively, the raw data can be provided based on the requirement for the publication of paper. The required data for the research paper can be provided by the corresponding author (B.Amalore Naveen Antony) based on the intent of the publisher.

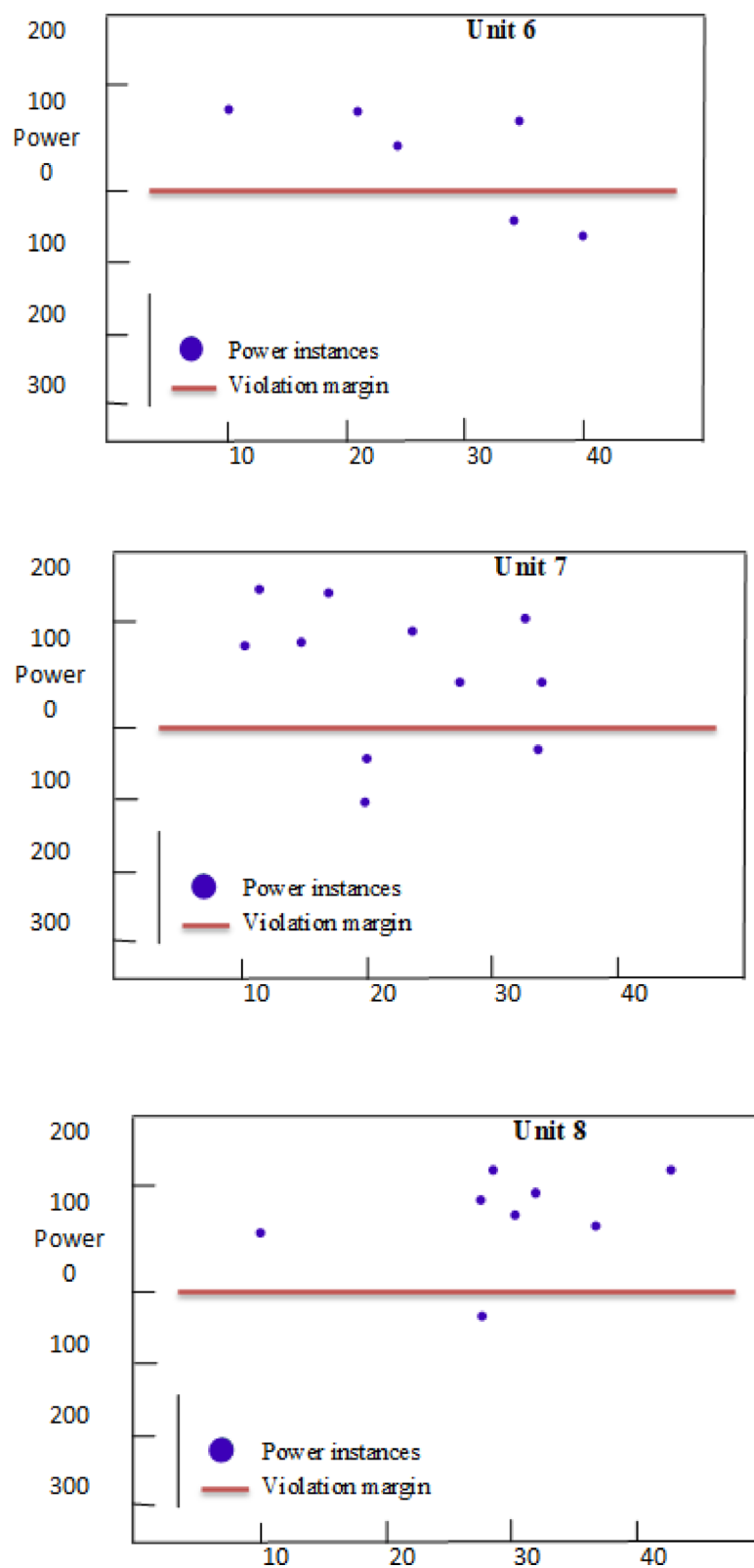


Fig. 10. (continued)

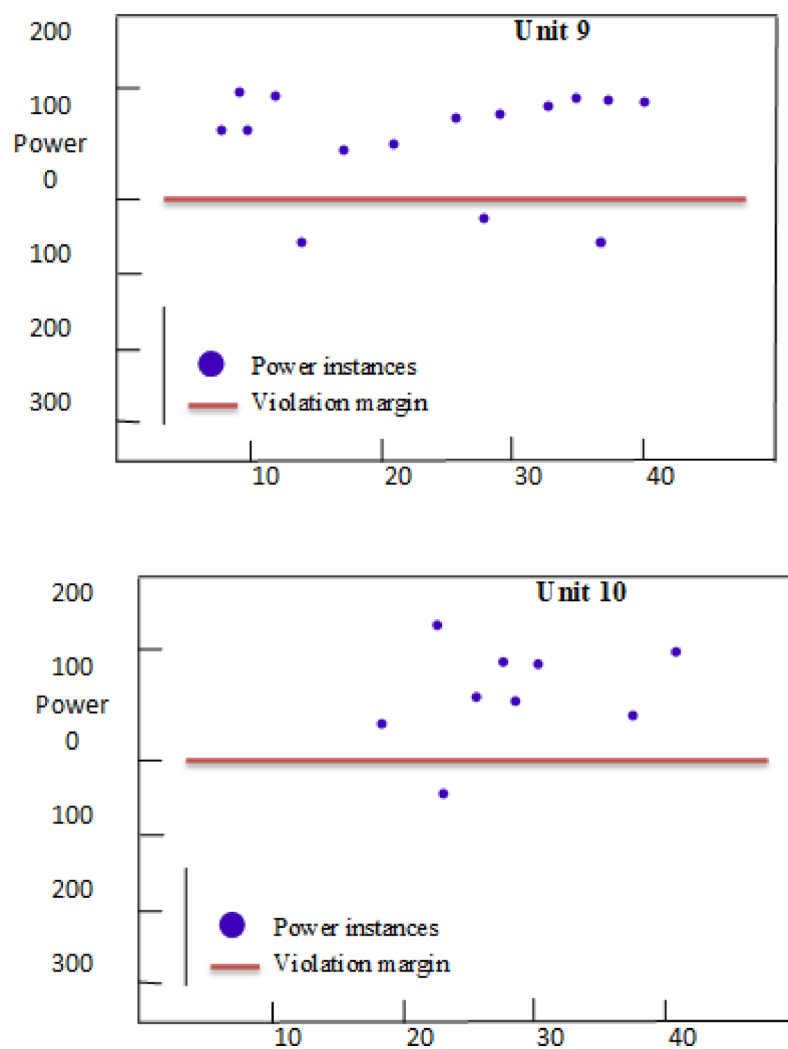


Fig. 10. (continued)

Station details	Deviation of power (in MW)	Conventional ABT mechanism		Penalty indicator programme method	
		Deviation rate in percentage	Penalty rate in Paise/KWh	Deviation strike rate per day	Penalty rate as per PIP formulation in Rs/kwh
Singrauli	45	3%	174	12	261
Rihand I&II	13	0.9%	-	08	174
Unchahar	-6	0.7%	-	07	174
Dadri-I&II	83	0.6%	-	07	174
Jhajjar	0	-	-	-	-
Shree cement	-70	26%	435	35	435
Dadri gas	10	0.1%	-	03	-
Auraiya gas	4	0.1%	-	03	-
Anta gas	5	0.1%	-	-	-
Narora	-22	0.8%	-	07	174
Nathpa jhakri	-96	19%	435	10	174
Bhakra	-19	0.2%	-	04	-
Tehri	-7	0.9%	-	06	174
Dehar	-27	0.4%	-	04	-
Chamera I	-9	0.2%	-	05	174
Salal I&II	-59	1%	-	09	174
Uri	2	0.4%	-	04	-
Pong	0	0	-	-	-
Dulhastrri	0	0	-	-	-
Dhauliganga	-2	0.7%	-	08	174
Chamera	0	-	-	-	-
Tanakpur	1	1%	-	10	174

Table 4. Comparison of penalty Estimation based on existing ABT and proposed PIP technique.

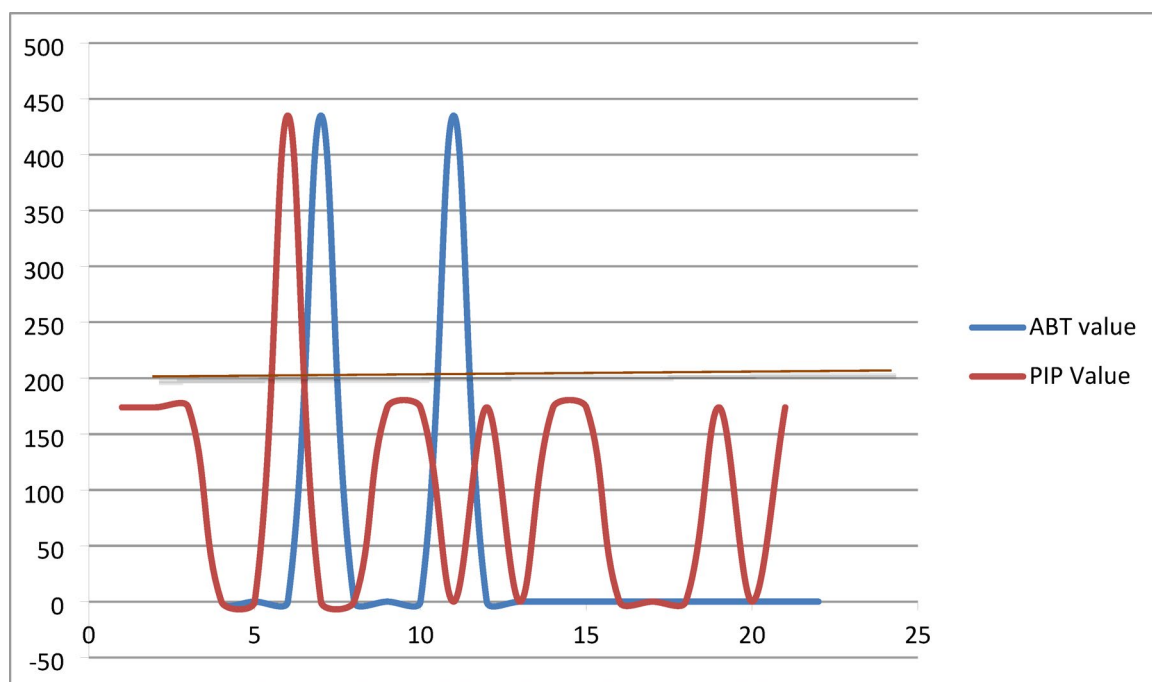


Fig. 11. Comparison between penalty estimation between existing ABT and proposed PIP method.

Data availability

As the power outage statistical data's and the cost evaluation for the unscheduled interchange between the power lines have been stated based on the report of the enquiry committee on grid disturbance submitted to the union government of India and the amendments formulated by central electricity regulatory commission respectively, the raw data can be provided based on the requirement for the publication of paper. The required data for the research paper can be provided by the corresponding author (B.Amalore Naveen Antony), aligning with the publisher's requirements.

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Author contributions

Mr.B.Amalore conceived the research area and arrange the flow of research article Dr.V.Mohan has analyze the statments and collecting evidences and statisitcal proofs for the article Mrs.G.Chitrakala has provided software knowledge for formulating the solution technique.

Declarations

Competing interests

The authors declare no competing interests.

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